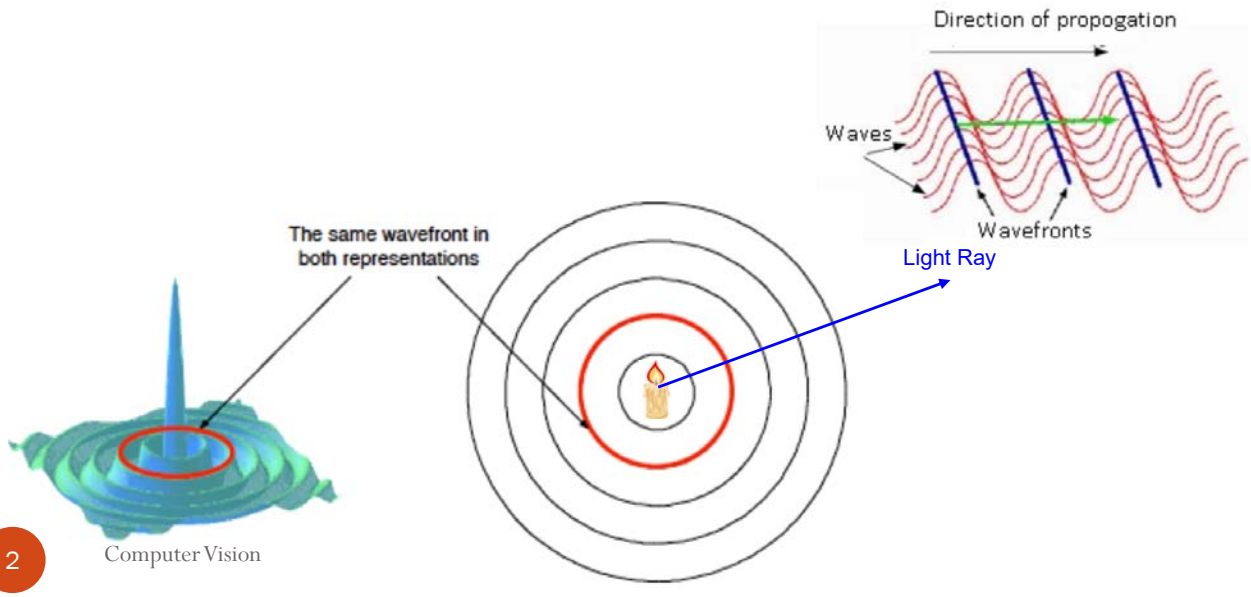
Computer Vision
Lecture 5: Optics

Yeungnam University
MSP lab.
Ho-Youl JUNG

영남대학교 Yeungnam University

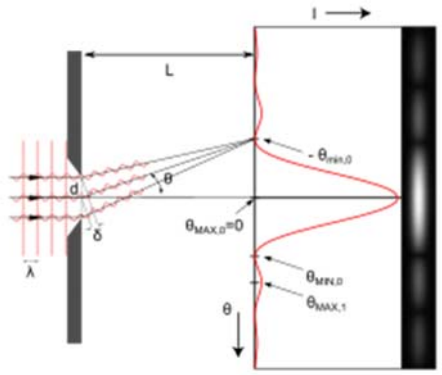
Wavefronts(파면) and Rays(광선)

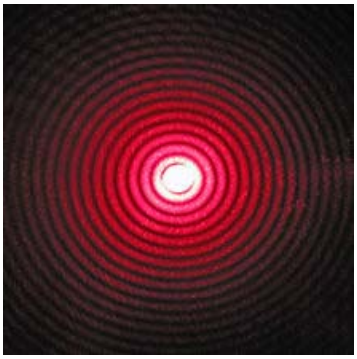
- Light propagates from a point source in spherical waves
- Wavefront is a surface where all rays from a source point have the same phase
- The path of pointing a given point on the wavefront is called ray

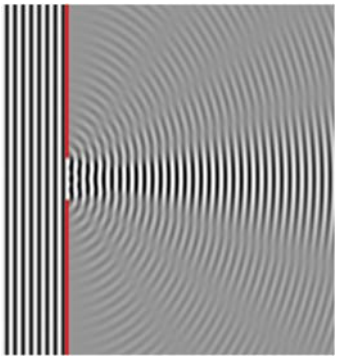


Diffraction(회절)

- **Diffraction** refers to various phenomena that occur when a wave encounters an obstacle or opening
- Defined as the bending of waves around the corners of an obstacle or through an aperture into the region of geometrical shadow of the obstacle/aperture

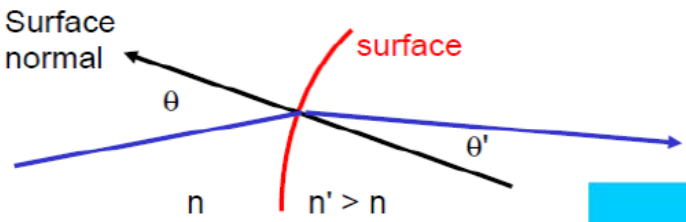


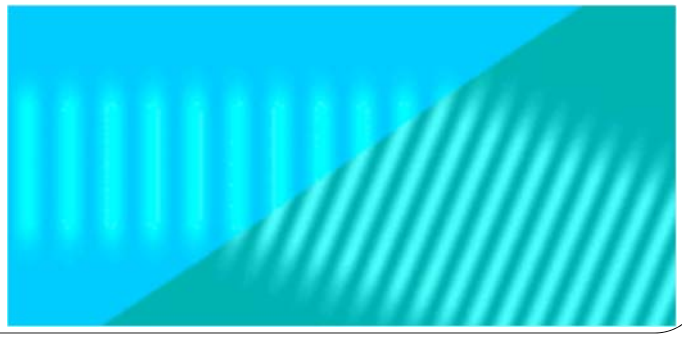




Refraction(굴절), Refractive Index(굴절률)

- When light enters an medium other than vacuum, it slow down
- The ratio of the speed of light in vacuum to that in the medium is called the index of refraction :
  - $n = \frac{c}{v}$, c: $speed\ of\ light\ in\ vacuum$, v: $speed\ of\ light\ in\ medium$
  - Ex: visible glass ➔ n =1.5~2.0

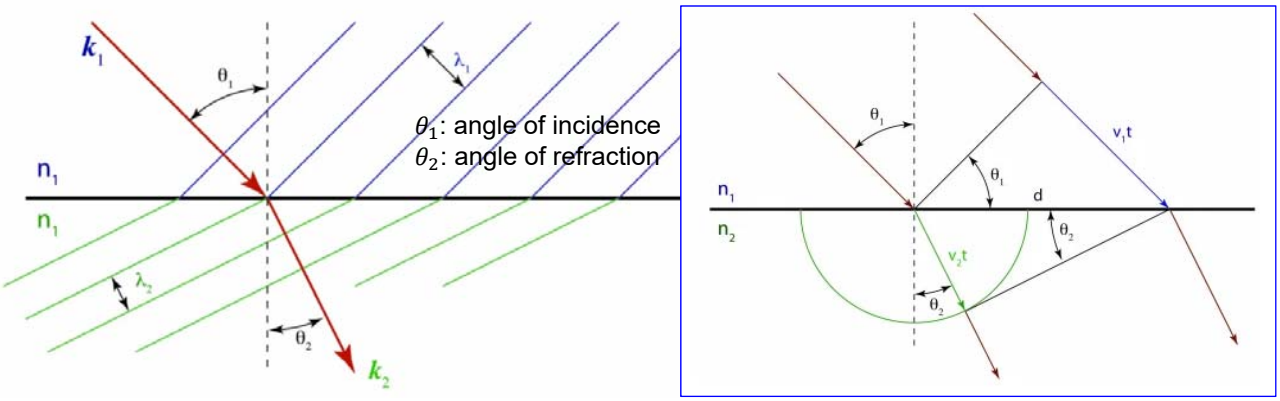




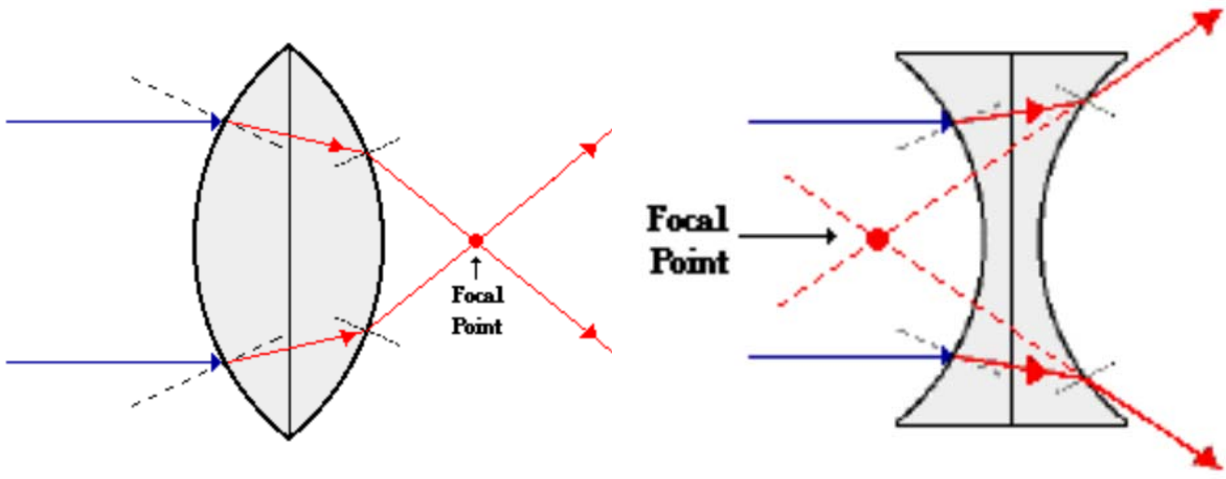
Snell' s Law (Law of Refraction)

- Refraction at any surface follows Snell's Law
  - $\lambda_1 = v_1 t,\ \sin\theta_1 = \frac{\lambda_1}{d} = \frac{v_1 t}{d}\ ,\ n_1 = \frac{c}{v_1}$ ➔ $\sin\theta_1 = \frac{c}{n_1}\frac{t}{d}$
  - $\lambda_2 = v_2 t,\ \sin\theta_2 = \frac{\lambda_2}{d} = \frac{v_2 t}{d}\ ,\ n_2 = \frac{c}{v_2}$ ➔ $\sin\theta_2 = \frac{c}{n_2}\frac{t}{d}$

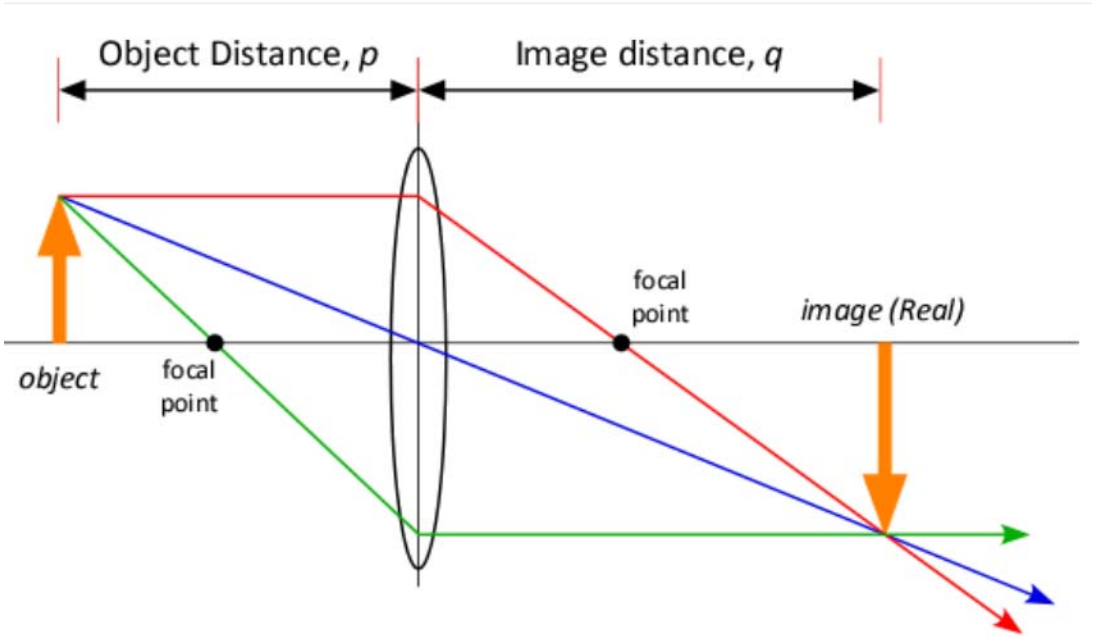
Snell's Law $n_1 \sin\theta_1 = n_2 \sin\theta_2$



Refraction by converging(convex) and diverging(concave) lenses



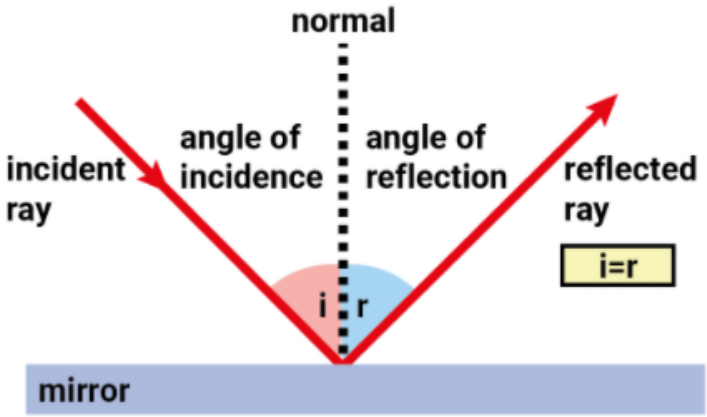
Converging(convex) lens



https://physics.stackexchange.com/questions/464866/determining-focal-length-of-converging-lens

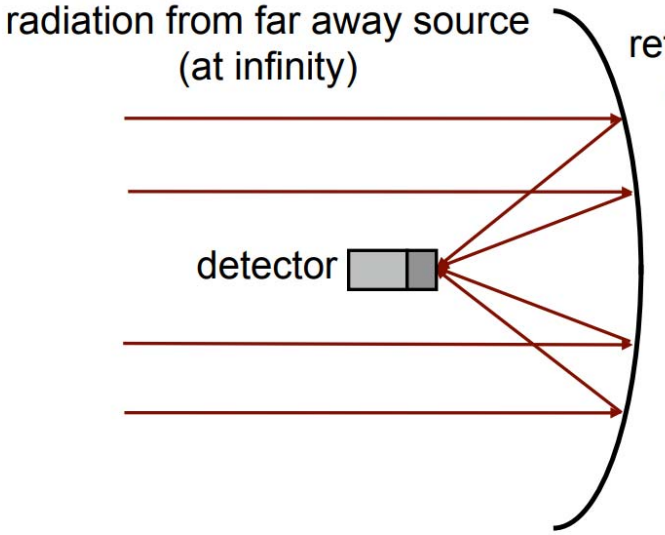
Reflection

- When a ray of light approaches a smooth polished surface and the light ray bounces back, it is called the reflection of light
- The ray that bounces back is called the reflected ray.
- The figure below shows the reflection of an incident beam on a plane mirror
- If $n_1 = 1$ and $n_2 = -1$ in Snell's Law, sin(i)=sin(r)
  - r=-I ➔ The angle of incident equals of reflection one



https://www.toppr.com/ask/content/story/amp/laws-of-reflection-49159/

Parabola reflector/antenna





Courtesy of NASA/JPL-Caltech.

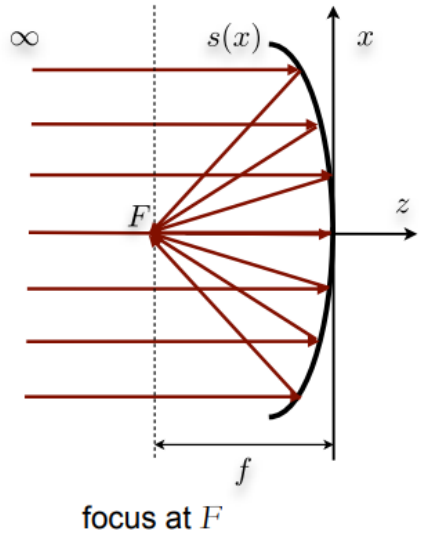


Image by hyperborea at Flickr.

Applications:
solar concentrators, satellite dishes, radio telescopes

https://ocw.mit.edu/courses/mechanical-engineering/2-71-optics-spring-2009/video-lectures/lecture-3-focusing-imaging-and-the-paraxial-approximation/MIT2_71S09_lec03.pdf

Parabola reflector/antenna : perfect focusing



focus at F

What should the shape function $s(x)$ be in order for the incoming parallel ray bundle to come to perfect focus?

The easiest way to find the answer is to invoke Fermat's principle: since the rays from infinity follow the *minimum* path before they meet at P, it follows that they must follow the *same* path.

$$2f = f - s + \sqrt{x^2 + (f-s)^2}$$

$$f + s = \sqrt{x^2 + (f-s)^2}$$

$$\begin{aligned} x^2 &= (f+s)^2 - (f-s)^2 \Rightarrow \\ &= 4sf \Rightarrow \end{aligned}$$

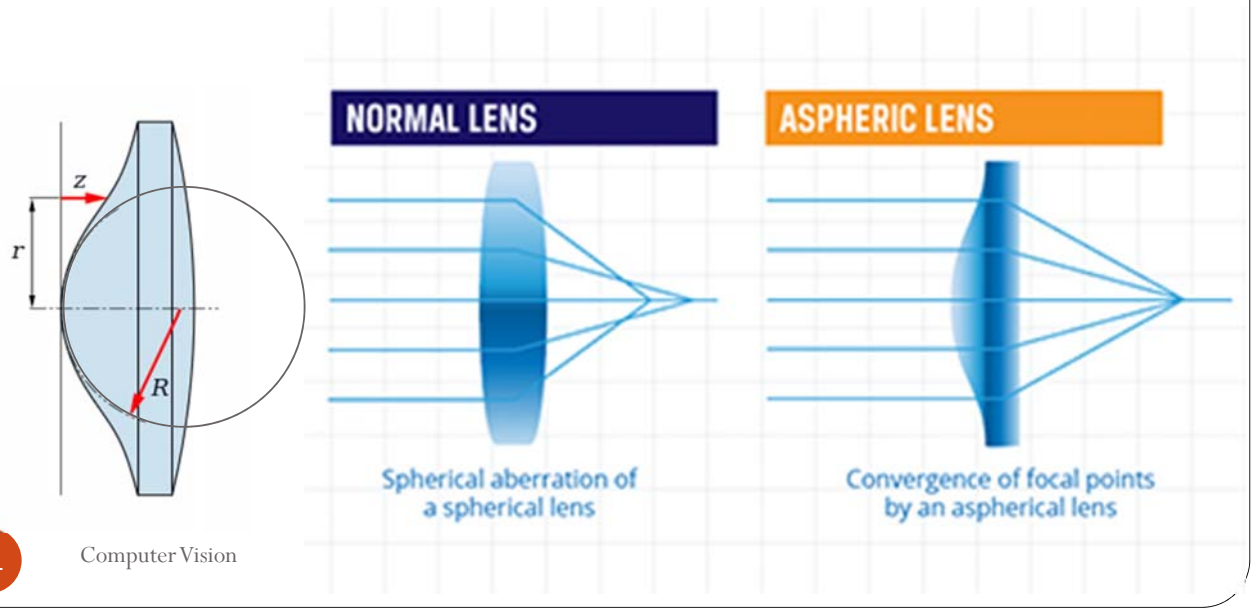
A paraboloidal reflector focuses a normally incident plane wave to a point

$$\boxed{s(x) = \frac{x^2}{4f}}$$

https://ocw.mit.edu/courses/mechanical-engineering/2-71-optics-spring-2009/video-lectures/lecture-3-focusing-imaging-and-the-paraxial-approximation/MIT2_71S09_lec03.pdf

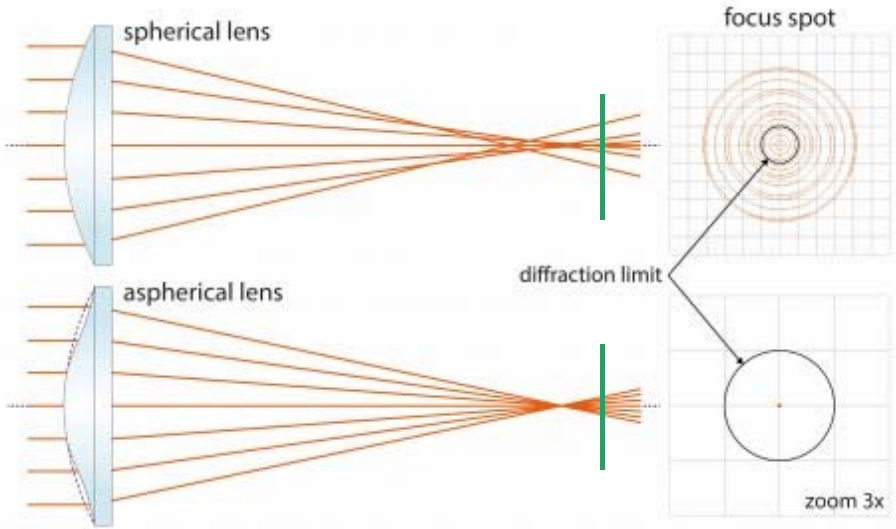
Spherical and aspherical lens

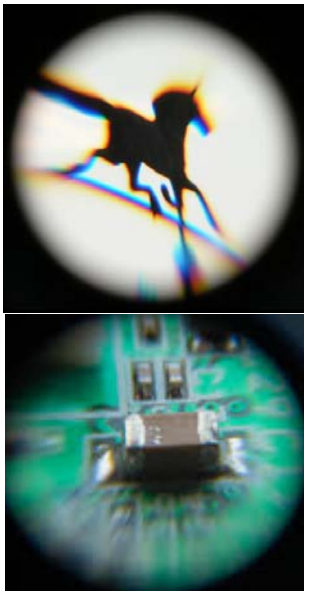
- **Normal (Spherical) lens**
- **Aspheric lens** is a lens whose surface profiles are not portions of a sphere or cylinder



Spherical aberration (구면수차/변형)

- **Spherical aberrations** occur in systems with spherical surfaces
- Spherical surfaces reflect or refract incident light rays stronger at margins than close to the optical axis
- The result are widespread focus sports or blurred images as no distinct focus is formed

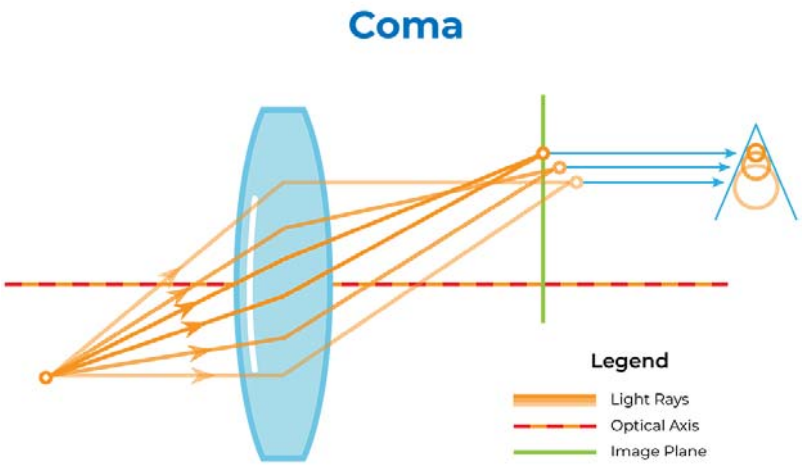


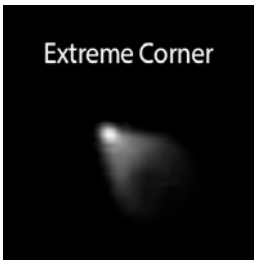


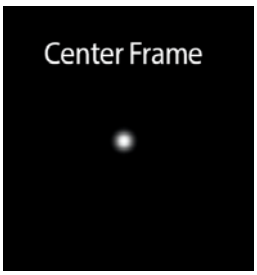
https://www.novuslight.com/uploads/n/illustration-spherical-aberration-V2.jpg
https://www.rvcj.com/wp-content/uploads/2016/11/Spherical-Aberration.png

Coma

- Coma (known as “comatic aberration”) is a type of optical aberration that results in off-axis points of light appearing comet-shaped
- As light rays from the edges of the frame pass through various parts of a spherical surface, they vary in magnification, creating a series of asymmetrical circular shapes of increasing sizes
- Unlike chromatic aberration, coma damages an image permanently, and it cannot be fixed via post-processing software



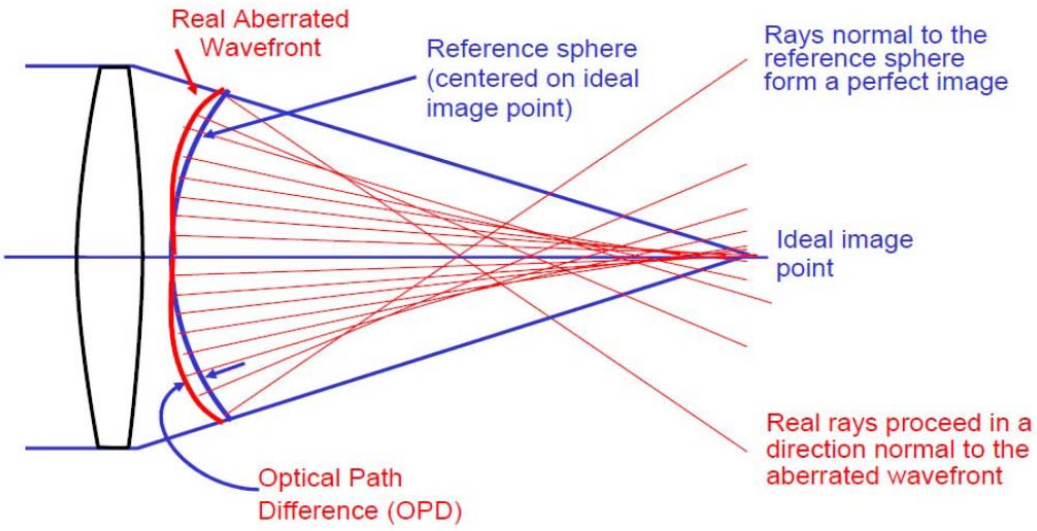




https://photographylife.com/wp-content/uploads/2019/12/Coma-Optics.png

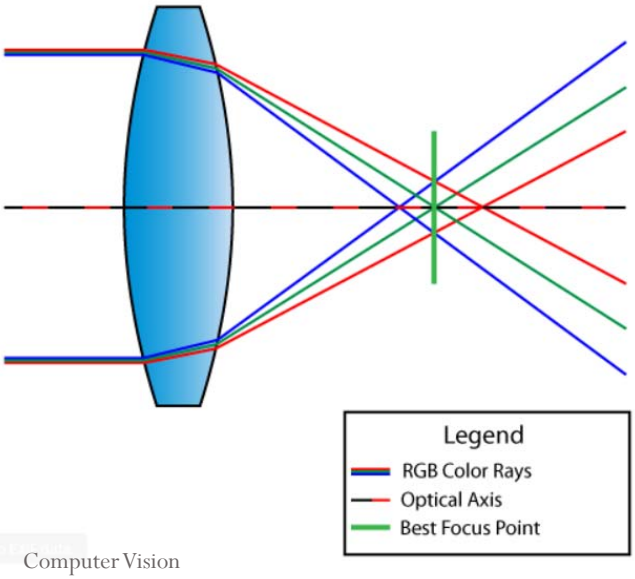
Spherical aberration(구면수차/변형)

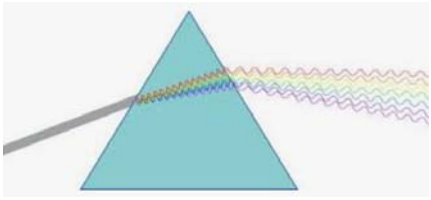
- **Spherical aberrations** occur when the wavefront is not perfect spherical

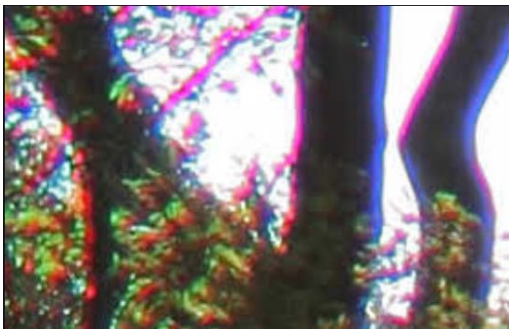


Chromatic aberration (색수차)

- **Chromatic aberrations** occur by the lens's refractive index changing with wavelength
- The shorter wavelengths focus closer to the lens because it has higher refractive index





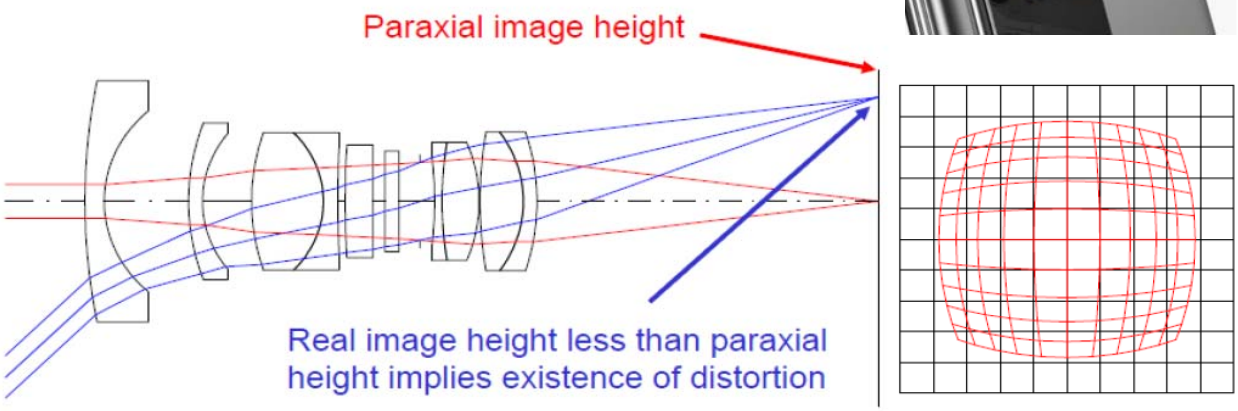


https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcSbSqDZdoqcXC-0kcGlc7DbWjdGsL6yUSkIQg&usqp=CAU

Radial Distortion

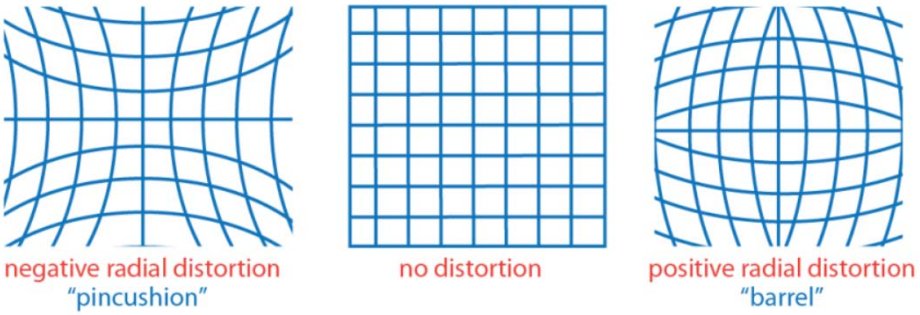
- **Distortion** is particularly associated with zoom lens, particularly large-range zooms and depends on focal distance
- **Barrel distortion** is often seen in wide-angle lenses
- **Pincushion distortion** is often seen in long focus lenses





Radial Distortion

- **Barrel distortion** vs **Pincushion distortion**

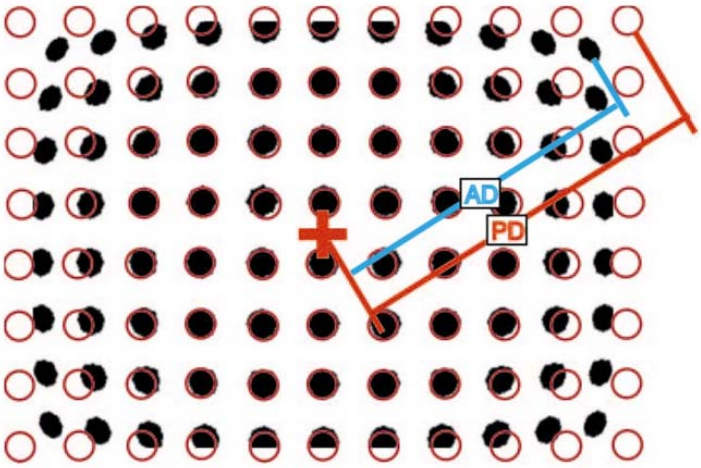






https://www.researchgate.net/figure/Radial-lens-distortion-figure-taken-from-aumathworkscom-help-vision-ref_fig4_328900961
https://forum.doom9.org/showthread.php?p=1396243

Radial Distortion



- % distortion : $= \frac{(AD-PD)}{PD} \times 100\ \%$
- AD is Actual Distance that an image point is from center of the field
- PD is the Predicted Distance that an image point would be from the center of the field if no distortion were present

Thank You
Next : Camera structure

